

LEWIS STRUCTURES WORKSHEET**Determining Bond Type:**

Use your knowledge of electronegativity to determine whether bonds within the following molecules would be covalent, polar-covalent, or ionic.

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|--------------------------|----------------------------|
| a) Na_2O | d) Al_2O_3 |
| b) RbBr | e) NF_3 |
| c) Cl_2 | f) SCl_4 |

Ionic Bonds

Each of the following molecules contain an ionic bond between atoms.

i) Draw the Lewis structures for each molecule.

- | | |
|-------------------------|--------------------------|
| a) MgS | f) MgI_2 |
| b) CaF_2 | g) Li_3P |
| c) KI | h) SrBr_2 |
| d) LiBr | i) BaCl_2 |
| e) K_2O | j) K_2S |

Covalent Bonds:**PART A**

Each of the following molecules contain only single bonds between atoms. .

i) Draw the Lewis Structure for each molecule.

- | | | |
|---------------------------|--------------------------|----------------------------|
| a) Cl_2 | f) HCl | k) CHCl_3 |
| b) ICl | g) CH_4 | l) S_2Cl_2 |
| c) H_2O | h) NF_3 | m) Br_2 |
| d) PH_3 | i) Cl_2O | n) SiF_4 |
| e) H_2S_2 | j) PCl_3 | |

PART B

Each of the following molecules may contain single, double or triple bonds..

i) Draw the Lewis Structure for each molecule.

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|----------------------------|---------------------------|
| 1) CS_2 | 4) I_2 |
| 2) C_2Br_2 | 5) O_2 |
| 3) BrCl | 6) N_2H_2 |